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Burj Khalifa

The **Burj Khalifa**[a] (known as the **Burj Dubai** prior to its inauguration) is a skyscraper in Dubai, United Arab Emirates. It is the world's tallest structure. With a total height of 829.8 m (2,722 ft, or just over half a mile) and a roof height (excluding antenna, but including a 242.6 m spire)[2] of 828 m (2,717 ft), the Burj Khalifa has been the tallest structure and building in the world since its topping out in 2009, surpassing Taipei 101, the previous holder of that status.[3][4]

Construction of the Burj Khalifa began in 2004, with the exterior completed five years later in 2009. The primary structure is reinforced concrete and some of the structural steel for the building originated from the Palace of the Republic in East Berlin, the former East German parliament.[5] The building was opened in 2010 as part of a new development called Downtown Dubai. It was designed to be the centerpiece of large-scale, mixed-use development.

The building is named for the former president of the United Arab Emirates (UAE), Sheikh Khalifa bin Zayed Al Nahyan.[6] The United Arab Emirates government provided Dubai with financial support as the developer, Emaar Properties, experienced financial problems during the 2007-2008 financial crisis. Then president of the United Arab Emirates, Khalifa bin Zayed, organized federal financial support. For his support, Mohammad bin Rashid, Ruler of Dubai, changed the name from "Burj Dubai" to "Burj Khalifa" during inauguration.

The design is derived from the Islamic architecture of the region, such as in the Great Mosque of Samarra. The Y-shaped tripartite floor geometry is designed to optimise residential and hotel space. A buttressed central core and wings are used to support the height of the building. Although this design was derived from Tower Palace III, the Burj Khalifa's central core houses all vertical transportation except egress stairs within each of the wings.[7] The structure also features a cladding system which is designed to withstand Dubai's hot summer temperatures. [8] It contains a total of 57 elevators and 8 escalators.

Burj Khalifa	
برج خليفة	
Viewed across The Dubai Fountain	
Record height	
Tallest in the world since 2009[l]	
Preceded by	Taipei 101
General information	
Status	Completed
Type	Mixed-use
Architectural style	Neo-futurism
Location	Dubai, United Arab Emirates
Address	1 Sheikh Mohammed bin Rashid Boulevard
Named for	Sheikh Khalifa
Construction started	6 January 2004
Topped-out	17 January 2009
Completed	1 October 2009
Opened	4 January 2010
Cost	US$1.5 billion
Owner	Emaar Properties
Height	
Architectural	828 m (2,717 ft)
Tip	829.8 m (2,722 ft)
Antenna spire	242.5 m (796 ft)
Roof	739.4 m (2,426 ft)
Top floor	585.4 m (1,921 ft)
Observatory	555.7 m (1,823 ft)
Technical details	
Structural system	Reinforced concrete, steel, and aluminium
Floor count	154 + 9 maintenance
Floor area	309,473 m^2 (3,331,100 sq ft)
Lifts/elevators	57 (mall included)
Design and construction	
Architect(s)	Adrian Smith
Architecture firm	Skidmore, Owings & Merrill
Structural engineer	Bill Baker
Main contractor	Samsung C&T BESIX
Other information	
Parking	2 subterranean levels
Public transit access	M1 At Burj Khalifa/Dubai Mall
Website	
burjkhalifa.ae (http://burjkhalifa.ae)	
References	
[1]	

Development

Construction began on 12 January 2004, with the exterior of the structure completed on 1 October 2009. The building officially opened on 4 January 2010[9][10] and is part of the 2 km^2 (490-acre) Downtown Dubai development at the 'First Interchange' along Sheikh Zayed Road, near Dubai's main business district.

The tower's architecture and engineering were performed by Skidmore, Owings & Merrill of Chicago, with Adrian Smith as chief architect, and Bill Baker as a chief structural engineer.[11][12] The firm had designed the Sears Tower in Chicago, a previous record holder for the world's tallest building.

Hyder Consulting was supervising engineer and NORR Group Consultants supervised the architecture The primary contractor was Samsung C&T of South Korea, together with the Belgian group BESIX and the local company Arabtec.[13]

Numerous complaints concerned migrant workers from South Asia, the primary building labour force, who were paid low wages and sometimes had their passports confiscated.[14]

Conception

Burj Khalifa was designed to be the centerpiece of a large-scale, mixed-use development to include 30,000 homes,[15] nine hotels (including The Address Downtown Dubai), 3 hectares (7.4 acres) of parkland, at least 19 residential skyscrapers, the Dubai Mall, and the 12-hectare (30-acre) artificial Burj Khalifa Lake. The decision to build Burj Khalifa was reportedly based on the government's decision to diversify from an oil-based economy to one that is service and tourism based. According to officials, projects like Burj Khalifa needed to be built to garner more international recognition and hence investment. "He (Sheikh Mohammed bin Rashid Al Maktoum) wanted to put Dubai on the map with something really sensational," said Jacqui Josephson, a tourism and VIP delegations executive at Nakheel Properties.[16]

The tower was known as Burj Dubai ("Dubai Tower") until its official opening in January 2010.[17] It was renamed in honour of the ruler of Abu Dhabi, Khalifa bin Zayed Al Nahyan; Abu Dhabi and the federal government of UAE lent Dubai tens of billions of US dollars so that Dubai could pay its debts – Dubai borrowed at least $80 billion for construction projects.[17] In the 2000s, Dubai started diversifying its economy but it suffered from an economic crisis in 2007–2010, leaving large-scale projects already in construction abandoned.[18]

Records

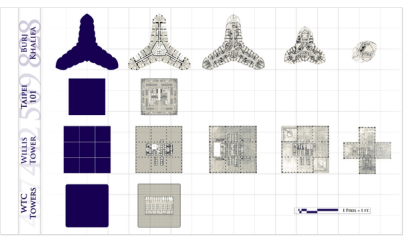
The Burj Khalifa set several world records, including:

- Tallest existing structure: 829.8 m (2,722 ft) (previously KVLY-TV mast – 628.8 m or 2,063 ft)
- Tallest structure ever built: 829.8 m (2,722 ft) (previously Warsaw radio mast – 646.38 m or 2,121 ft)
- Tallest freestanding structure: 829.8 m (2,722 ft) (previously CN Tower – 553.3 m or 1,815 ft)
- Tallest skyscraper (to top of spire): 828 m (2,717 ft) (previously Taipei 101 – 509.2 m or 1,671 ft)[19]
- Tallest skyscraper to top of antenna: 829.8 m (2,722 ft) (previously the Willis (formerly Sears) Tower – 527 m or 1,729 ft)
- Building with most floors: 163 (previously World Trade Center – 110)[20]
- World's highest elevator installation (situated inside a rod at the very top of the building)[21]
- World's longest travel distance elevators: 504 m (1,654 ft)[21][22]
- Highest vertical concrete pumping (for a building): 606 m (1,988 ft)[23]
- World's tallest structure that includes residential space[24]
- World's highest installation of an aluminium and glass façade: 512 m (1,680 ft)[25]
- World's highest restaurant (*At.mosphere*): 122nd floor at 442 m (1,450 ft) (previously 360, at a height of 350 m

(1,148 ft) in CN Tower)[26]

- World's highest New Year display of fireworks.[27]
- World's largest light and sound show staged on a single building.[28][29][30]

Architecture and design



Cross-section comparisons of various towers, from ground level to top: Burj Khalifa, Taipei 101, Willis Tower, and World Trade Center

The tower was designed by Skidmore, Owings, and Merrill (SOM), which also designed the Willis Tower (formerly the Sears Tower) in Chicago and the One World Trade Center in New York City. Burj Khalifa uses the bundled tube design of the Willis Tower, invented by Fazlur Rahman Khan.[31][32] Due to its tubular system, proportionally only half the amount of steel was used in the construction, compared to the Empire State Building.[31][33] Khan's contributions to the design of tall buildings have had a profound impact on architecture and engineering. It would be difficult to find any worldwide practices in the design of tall buildings that have not been directly or indirectly influenced by his work.[34] The design is reminiscent of Frank Lloyd Wright's vision for The Illinois, a mile-high skyscraper designed for Chicago, as well as Chicago's Lake Point Tower. When Adrian Smith was conceiving the project at SOM, he looked out his office window toward Lake Point Tower's curved three-wing layout and thought, "There's the prototype".[35] According to Strabala, Burj Khalifa was designed based on the 73rd floor Tower Palace Three, an all-residential building in Seoul. In its early planning, Burj Khalifa was intended to be entirely residential.[36]

After the original design by Skidmore, Owings, and Merrill, Emaar Properties chose Hyder Consulting to be the supervising engineer and NORR Group Consultants International Ltd to supervise the architecture of the project.[37] Hyder was selected for their expertise in structural and MEP (mechanical, electrical and plumbing) engineering.[38] Hyder Consulting's role was to supervise construction, certify the architect's design, and be the engineer and architect of record to the UAE authorities.[37] NORR's role was the supervision of all architectural components including on-site supervision during the construction and design of a 6-storey addition to the office annex building for architectural documentation. NORR was also responsible for the architectural integration drawings for the Armani Hotel included in the Tower. Emaar Properties also engaged GHD,[39] an international multidisciplinary consulting firm, to act as an independent verification and testing authority for concrete and steelwork.

The design is derived from Islamic architecture.[21] As the tower rises from the flat desert base, there are 27 setbacks in a spiral pattern, decreasing the cross-section of the tower as it rises and creating convenient outdoor terraces. These setbacks are arranged and aligned in a way that minimizes vibration wind loading from eddy currents and vortices.[7] At the top, the central core emerges and is sculpted to form a finishing spire. At its tallest point, the tower sways a total of 1.5 m (4.9 ft).[40]

The spiral minaret at the Great Mosque of Samarra

The spire of Burj Khalifa is composed of more than 4,000 tonnes (4,400 short tons; 3,900 long tons) of structural steel. The central pinnacle pipe weighs 350 tonnes (390 short tons; 340 long tons) and has a height of 200 m (660 ft). The spire also houses communications equipment.[41] This 244-metre (801 ft) spire is widely considered vanity height, since very little of its space is usable. Without the spire, Burj Khalifa would be 585 metres (1,919 ft) tall. This was reported in a Council on Tall Buildings and Urban Habitat study, which notes that the empty spire "could be a skyscraper on its own".[2] Such a skyscraper, if located in Europe, would be the 11th tallest building on that continent.[42]

In 2009 architects announced that more than 1,000 pieces of art would adorn the interiors of Burj Khalifa, while the residential lobby of Burj Khalifa would display the work of Jaume Plensa.[43]

The cladding system consists of 142,000 m^2 (1,528,000 sq ft) of more than 26,000 reflective glass panels and aluminium and textured stainless steel spandrel panels with vertical tubular fins.[41] The architectural glass provides solar and thermal performance as well as an anti-glare shield for the intense desert sun, extreme desert temperatures and strong winds. The glass covers more than 174,000 m^2 (1,870,000 sq ft) in area. The Burj's typical curtain wall panels measure 4 ft 6 in (1.4 m) wide by 10 ft 8 in (3.3 m) high and weigh about 800 pounds (360 kg) each, with wider panels near the building's edges and taller ones near the top.[44]

The exterior temperature at the top of the building is thought to be 6 °C (11 °F) cooler than at its base.[45]

A 304-room Armani Hotel, the first of 4 by Armani, occupies 15 of the lower 39 floors.[3][46] The hotel was supposed to open on 18 March 2010,[47][48] but after several delays, it finally opened to the public on 27 April 2010.[49] The corporate suites and offices were also supposed to open from March onwards,[50] yet the hotel and observation deck remained the only parts of the building which were open in April 2010.

The sky lobbies on the 43rd and 76th floors house swimming pools.[51] Floors 20 through 108 have 900 private residential apartments (which, according to the developer, sold out within eight hours of being on the market). An outdoor zero-entry swimming pool is located on the 76th floor of the tower. Corporate offices and suites fill most of the remaining floors, except for the 122nd, 123rd, and 124th, where the Atmosphere restaurant, sky lobby, and an indoor and outdoor observation deck are located respectively. In January 2010, it was planned that Burj Khalifa would receive its first residents in February 2010.[51][52]

The building has 57 elevators and 8 escalators.[41] The elevators have a capacity of 12 to 14 people per cabin, and include the world's fastest double-deck elevators, rising and descending at up to 10 m/s (33 ft/s). Engineers initially considered installing the world's first triple-deck elevators.[24] The double-deckers are equipped with LCD displays to amuse visitors during their travel to the observation deck.[53] The building has 2,909 stairs from the ground floor to the 160th floor.[54]

Plumbing systems

The Burj Khalifa's water system supplies an average of 946,000 L (250,000 U.S. gal) of water per day through 100 km (62 mi) of pipes.[21][55] An additional 213 km (132 mi) of piping serves the fire emergency system, and 34 km (21 mi) supplies chilled water for the air conditioning system.[55]

Air conditioning

The air conditioning system draws air from the upper floors where the air is cooler and cleaner than on the ground.[56] At peak cooling times, the tower's cooling is 46 MW (62,000 hp), equivalent to that provided by 13,000 short tons (26,000,000 lb; 12,000,000 kg) of melting ice in one day.[55] Water is collected via a condensate collection system and is used to irrigate the nearby park.[21]

Window cleaning

To wash the 24,348 windows, totaling 120,000 m^2 (1,290,000 sq ft) of glass, the building has three horizontal tracks, each holding a 1,500 kg (3,300 lb) bucket machine. Above level 109, and up to tier 27, traditional cradles from davits are used. The top of the building is cleaned by a crew that uses ropes to descend from the top to gain access.[57][58] Under normal conditions, when all building maintenance units are operational, it takes 36 workers three to four months to clean the entire exterior.[41][59]

Unmanned machines clean the top 27 additional tiers and the glass spire. The cleaning system was developed in Melbourne, Australia, by CoxGomyl, a manufacturer of building maintenance units,[60] at a cost of A$8 million.[59]

Features

Fountain

The Dubai Fountain

Outside the Burj Khalifa, WET Enterprises designed a fountain system at a cost of Dh 800 million (US$217 million). Illuminated by 6,600 lights and 50 coloured projectors, it is 270 m (900 ft) long and shoots water 150 m (500 ft) into the air while accompanied by a range of classical to contemporary Arabic and other music. It is the world's largest choreographed fountain.[61] On 26 October 2008, Emaar announced that based on results of a naming contest the fountain would be called the Dubai Fountain.[62]

Observation deck

View of The Dubai Fountain from the observation deck

View from the observation deck

An outdoor observation deck, named *At the Top*, opened on 5 January 2010 on the 124th floor. At 452 m (1,483 ft), it was the highest outdoor observation deck in the world when it opened.[63] After this was initially surpassed in December 2011 by *Cloud Top 488* on the Canton Tower, Guangzhou at 488 m (1,601 ft),[64] Burj Khalifa opened the 148th floor SKY level at 555 m (1,821 ft), once again giving it the highest observation deck in the world on 15 October 2014. Subsequently, on February 18, 2019, the Burj Khalifa also opened The Lounge observatory at 584 m (1,916 ft) – the highest lounge in the world,[65][66] However, in June 2016 the Shanghai Tower opened with an observation deck at a height of 561 metres, thus taking the title of the world's highest observation deck.

The Burj Khalifa's 124th floor observation deck also features a so-called electronic telescope, an augmented reality device developed by Gsmprjct° of Montréal, which allows visitors to view the surrounding landscape in real-time, and to view previously saved images such as those taken at different times of day or under different weather conditions.[67][68][69] To reduce the daily rush of sightseers, management allows visitors to purchase tickets in advance for a specific date and time, at a 75% discount on tickets purchased on the spot.[70]

On 8 February 2010, the observation deck was closed to the public for two months after power-supply problems caused an elevator to become stuck between floors, trapping a group of tourists for 45 minutes.[71][72][73][74][75]

When the tide is low and visibility is high, people can see the shores of Iran (which is around 153 km or 95 mi away) from the top of the skyscraper.[76]

Park

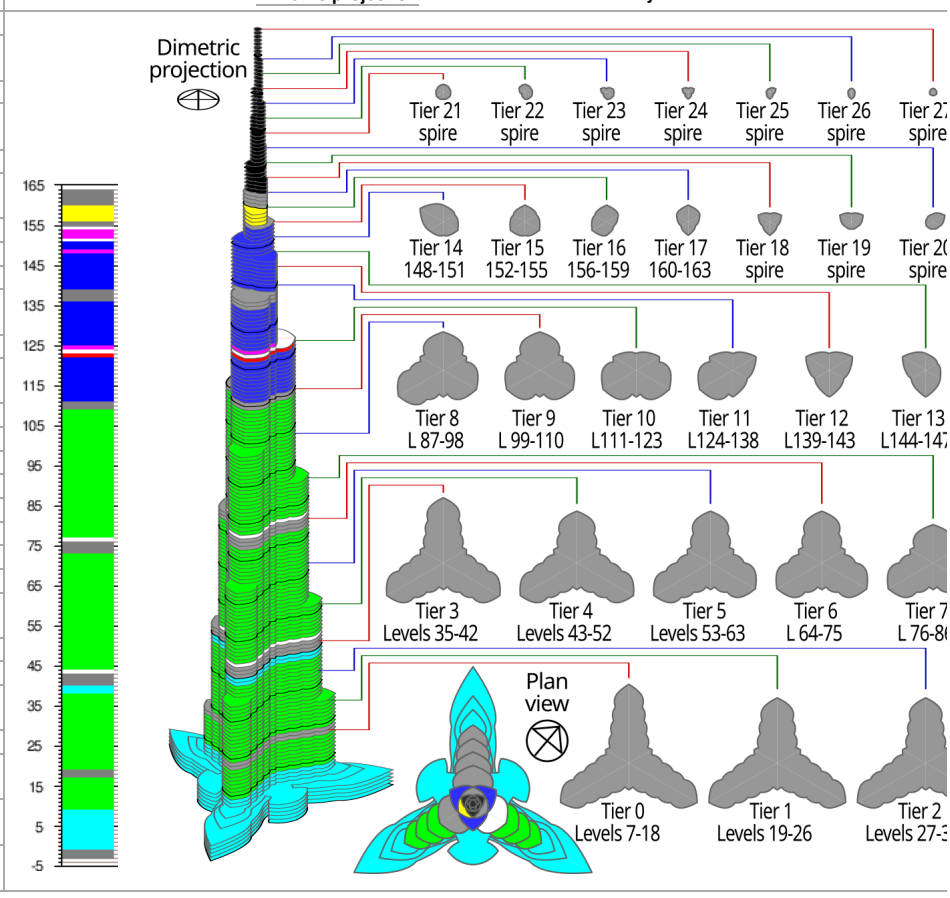
Burj Khalifa is surrounded by an eleven-hectare (27-acre) park designed by landscape architects SWA Group.[77] Like the tower, the park's design was based on the flower of the *Hymenocallis*, a desert plant.[78] At the centre of the park is the water room, which is a series of pools and water jet fountains. Benches and signs incorporate images of Burj Khalifa and the *Hymenocallis* flower.[79]

The plants are watered by water collected from the building's cooling system. The system provides 68,000,000 L (18,000,000 US gal) annually.[79] WET Enterprises, who also developed the Dubai Fountain, developed the park's six water features.[80]

Floor plan

Floors	Purpose[41][81]
160–163	Mechanical
156–159	Communication and broadcast
155	Mechanical
152-154	*The Lounge* observatory
149–151	Corporate suites
148	*At the Top Sky* observatory
139–147	Corporate suites
136–138	Mechanical
125–135	Corporate suites
124	*At the Top* observatory
123	Sky lobby
122	*At.mosphere* restaurant
111–121	Corporate suites
109–110	Mechanical
77–108	Residential
76	Sky lobby
73–75	Mechanical
44–72	Residential
43	Sky lobby
40–42	Mechanical
38–39	Armani Hotel suites
19–37	Residential
17–18	Mechanical
9–16	Armani Residences
1–8	Armani Hotel
Ground	Armani Hotel, Lobby
Concourse	Armani Hotel, Lobby
B1–B2	Parking, mechanical

Dimetric projection with floors colour-coded by function[82]



Ramadan observance

On the higher floors, the sun is seen for several minutes after it has set at ground level. Those living above the 80th floor should wait two extra minutes to break their Ramadan fast, and those living above the 150th floor should wait three minutes.[83]

Construction

Animation of construction process

Aerial closeup of Burj Khalifa under construction in March 2008

The tower was constructed by Samsung C&T from South Korea, which also did work on the Petronas Twin Towers and Taipei 101.[84] Samsung C&T built the tower in a joint venture with BESIX from Belgium and Arabtec from the UAE.[85][86] Turner was the project manager on the main construction contract.[87] Hong Kong-based Far East Aluminium combined to provide the exterior cladding for Burj Khalifa.[88][89]

The contractor and the engineer of record was Hyder Consulting.[90] Under UAE law, the contractor and the engineer of record is jointly and severally liable for the performance of Burj Khalifa.

The primary structure is reinforced concrete. Putzmeister created a new, super high-pressure trailer concrete pump, the BSA 14000 SHP-D, for this project.[23] Burj Khalifa's construction used 330,000 m^3 (431,600 cu yd) of concrete and 55,000 tonnes (61,000 short tons; 54,000 long tons) of steel rebar, and construction took 22 million man-hours.[11] In May 2008 Putzmeister pumped concrete with more than 21 MPA ultimate compressive strength of gravel to surpass the 600 metres weight of the effective area of each column from the foundation to the next 4th level, and the rest was by metal columns *jacketed* or *covered* with concrete to a then world record delivery height of 606 m (1,988 ft),[23] the 156th floor. Three tower cranes were used during the construction of the uppermost levels, each capable of lifting a 25-tonne load.[91] The remaining structure above was constructed of lighter steel.

In 2003, 33 test holes were drilled to study the strength of the bedrock underlying the structure.[92] *"Weak to very weak sandstone and siltstone"* was found, just metres below the surface.[93] Samples were taken from test holes drilled to a depth of 140 metres, finding weak to very weak rock all the way.[94] The study described the site as part of a "seismically active area". Another challenging element was the shamal which often creates sandstorms.[19]

Over 45,000 m^3 (58,900 cu yd) of concrete, weighing more than 110,000 tonnes (120,000 short tons; 110,000 long tons) were used to construct the concrete and steel foundation, which features 192 piles; each pile is 1.5 metre in diameter by 43 m in length, buried more than 50 m (164 ft) deep.[24] The foundation was designed to support the total building weight of approximately 450,000 tonnes (500,000 short tons; 440,000 long tons). This weight was then divided by the compressive strength of concrete which is 30 MPa which yielded 450 sq. metres of vertical normal effective area, which then yielded 12 metres by 12 metres dimensions.[95] A cathodic protection system is under the concrete to neutralise the sulphate and chloride-rich groundwater and prevent corrosion.[41][19]

During the construction of the Burj Khalifa, over 35,000 tonnes of structural steel which held the Palace of the Republic, the former parliament building of the German Democratic Republic, the Volkskammer, in East Berlin together were shipped to Dubai in 2008.[5]

The Burj Khalifa is highly compartmentalised. Pressurised, air-conditioned refuge floors are located every 13 floors (on floors G, 13, 26, 39, 52, etc.) where people can shelter on their long walk down to safety in case of an emergency or fire.[41][96]

Special mixes of concrete were made to withstand the extreme pressures of the massive building weight; as is typical with reinforced concrete construction, each batch of concrete was tested to ensure it could withstand certain pressures. CTLGroup, working for Skidmore, Owings and Merrill, conducted the creep and shrinkage testing critical for the structural analysis of the building.[97]

The consistency of the concrete used in the project was essential. It was difficult to create a concrete that could withstand both the thousands of tonnes bearing down on it and Persian Gulf temperatures that can reach 50 °C (122 °F). To combat this problem, the concrete was not poured during the day. Instead, during the summer months, ice was added to the mixture and it was poured at night when the air was cooler and the humidity was higher. Cooler concrete cures more evenly and is, therefore, less likely to set too quickly and crack. Any significant cracks could have put the entire project in jeopardy.[98][99][100][101][102]

Milestones

Burj Khalifa and skyline of Dubai, 2010

- January 2004: Excavation commences.[25]
- February 2004: Piling starts.[25]
- 21 September 2004: Emaar contractors begin construction.[103]
- March 2005: Structure of Burj Khalifa starts rising.[25]
- June 2006: Level 50 is reached.[25]
- February 2007: Surpasses the Sears Tower as the building with the most floors.
- 13 May 2007: Sets record for vertical concrete pumping on any building at 452 m (1,483 ft), surpassing the 449.2 m (1,474 ft) to which concrete was pumped during the construction of Taipei 101, while Burj Khalifa reached the 130th floor.[25][104]
- 21 July 2007: Surpasses Taipei 101, whose height of 509.2 m (1,671 ft) made it the world's tallest building, and level 141 reached.[25][105]
- 12 August 2007: Surpasses the Sears Tower antenna, which stands 527 m (1,729 ft).
- 12 September 2007: At 555.3 m (1,822 ft), becomes the world's tallest freestanding structure, surpassing the CN Tower in Toronto, and level 150 reached.[25][106]
- 7 April 2008: At 629 m (2,064 ft), surpasses the KVLY-TV Mast to become the tallest human-made structure, level 160 reached.[25][107]
- 17 June 2008: Emaar announces that Burj Khalifa's height is over 636 m (2,087 ft) and that its final height will not be given until it is completed in September 2009.[108]
- 1 September 2008: Height tops 688 m (2,257 ft), making it the tallest human-made structure ever built, surpassing the previous record-holder, the Warsaw Radio Mast in Konstantynów, Poland.[109]
- 17 January 2009: Topped out at 829.8 m (2,722 ft).[110]
- 1 October 2009: Emaar announces that the exterior of the building is completed.[111]
- 4 January 2010: Burj Khalifa's official launch ceremony is held and Burj Khalifa is opened. Burj Dubai was renamed Burj Khalifa in honour of the President of the UAE and ruler of Abu Dhabi, Sheikh Khalifa bin Zayed al Nahyan.[4]
- 10 March 2010: Council on Tall Buildings and Urban Habitat certifies Burj Khalifa as world's tallest building.[112]

Real estate values

In March 2009, Mohamed Ali Alabbar, chairman of the project's developer, Emaar Properties, said office space pricing at Burj Khalifa reached US$4,000 per sq ft (over US$43,000 per m^2) and the Armani Residences, also in Burj Khalifa, sold for US$3,500 per sq ft (over US$37,500 per m^2).[113] He estimated the total cost for the project to be about US$1.5 billion.[6]

The project's completion coincided with the financial crisis of 2007–2008, and with vast overbuilding in the country, leading to high vacancies and foreclosures.[114] With Dubai mired in debt from its huge ambitions, the government was forced to seek multibillion-dollar bailouts from its oil-rich neighbor Abu Dhabi. Subsequently, in a surprise move at its opening ceremony, the tower was renamed Burj Khalifa, said to honour the UAE President Khalifa bin Zayed Al Nahyan for his crucial support.[4][115]

Because of the slumping demand in Dubai's property market, the rents in the Burj Khalifa plummeted 40% some ten months after its opening. Out of 900 apartments in the tower, 825 were still empty at that time.[116][117] Over the next 30 months, overseas investors steadily bought up available apartments and office space.[118] By October 2012, Emaar reported that around 80% of the apartments were occupied.[119]

Official launch ceremony

The ceremony was broadcast live on a giant screen on Burj Park Island and on smaller screens elsewhere.[120] Hundreds of media outlets from around the world reported live from the scene.[121] In addition to the media presence, 6,000 guests were expected.[122]

The opening was held on 4 January 2010.[123] The ceremony featured a display of 10,000 fireworks, light beams projected on and around the tower, and further sound, light and water effects.[121] The celebratory lighting was designed by UK lighting designers Speirs and Major Associates.[124] Using the 868 powerful stroboscope lights that are integrated into the façade and spire of the tower, different lighting sequences were choreographed, together with more than 50 different combinations of other effects.

Controversies

Deadly incidents

On 10 May 2011, an Asian migrant worker in his mid-30s jumped to his death from the 147th floor onto the 108th floor's deck. Dubai police said he killed himself because his company refused to let him leave the country.[125]

On 18 May 2015, Dubai police disputed a report that a Portuguese tourist named Laura Vanessa Nunes fell to her death from the Burj Khalifa the prior 16 November, claiming that she fell from the Jumeirah Lake Towers.[126][127] Nine News obtained emails from Portugal's embassy in the UAE under freedom of information laws, which indicated that the female tourist jumped from the 148th floor of the Burj Khalifa.[128]

Labour

The Burj Khalifa was built primarily by workers from South Asia and East Asia.[129][130] This is generally because the current generation of UAE locals prefer governmental jobs and do not have an attitude favouring private sector employment.[131][132] On 17 June 2008, there were about 7,500 skilled workers employed at the construction site.[108] Press reports indicated in 2006 that skilled carpenters at the site earned £4.34 a day, and labourers earned £2.84.[129] According

to a BBC investigation and a Human Rights Watch report, the workers were housed in abysmal conditions, and worked long hours for low pay.[133][134][135] During construction, one construction-related death was reported.[136] Workplace injuries and deaths in the UAE are poorly documented, according to Human Rights Watch.[133]

In March 2006 about 2,500 workers, upset over buses that were delayed for the end of their shifts, protested and triggered a riot, damaging cars, offices, computers, and construction equipment.[129] A Dubai Interior Ministry official said the rioters caused almost £500,000 in damage.[129] Most of the workers involved in the riot returned the following day but refused to work.[129]

New Year's Eve

Emaar New Year's Eve is an annual event held every 31 December at Burj Khalifa, organised by Emaar Properties.[137][138] The event consists of fireworks launched from Burj Khalifa, a light and laser show on the facade of Burj Khalifa, and an accompanying soundtrack and a special fountain show on The Dubai Fountain choreographed to the soundtrack.[139][140] The Emaar New Year's Eve fireworks celebration originated in 2010 with the inauguration of the world's tallest building, Burj Khalifa. The celebration was broadcast live to more than 2 million people and lasted for 3 minutes.[141]

Since 2011, national live broadcasting rights have been held by Dubai Media Incorporated and Dubai TV.[141]

Emaar New Year's Eve has won two Guinness World Records, including 'Largest LED-Illuminated Facade' in 2015 and 2019.[142][143][144]

In 2017 and 2018, Emaar New Year's Eve was broadcast live on Twitter,[145][146] and YouTube. In 2020, it was broadcast live for the first time on Zoom.[147]

In 2021, Emaar celebrated togetherness in honour of frontline workers of the COVID-19 pandemic.[148]

From 2022 onwards, lasers are installed on Burj Khalifa each year that project animations onto The Dubai Fountain. The lasers are temporary for the event. [149]

BASE jumping

The building has been used by several experienced BASE jumpers for authorised and unauthorised BASE jumping:

In May 2008, Hervé Le Gallou and David McDonnell, dressed as engineers, entered Burj Khalifa (around 650 metres (2,130 ft) at the time), and jumped off a balcony situated several floors below the 160th floor.[150][151]

On 8 January 2010, with permission of the authorities, Nasr Al Niyadi and Omar Al Hegelan, from the Emirates Aviation Society, broke the world record for the highest BASE jump from a building after they leapt from a crane-suspended platform attached to the 160th floor at 672 m (2,205 ft). The two men descended the vertical drop at a speed of up to 220 km/h (140 mph), with enough time to open their parachutes 10 seconds into the 90-second jump.[152][153]

On 21 April 2014, with permission of the authorities and support from several sponsors, highly experienced French BASE jumpers Vince Reffet and Fred Fugen broke the Guinness world record for the highest BASE jump from a building after they leapt from a specially designed platform, built at the very top of the pinnacle, at 828 metres (2,717 feet).[154][155][156]

Climbing

On 28 March 2011, Alain "Spiderman" Robert scaled the outside of Burj Khalifa. The climb to the top of the spire took 6 hours. To comply with UAE safety laws, Robert, who usually climbs in free solo style, used a rope and harness.[157]

Awards

In June 2010, Burj Khalifa was the recipient of the 2010 "Best Tall Building Middle East & Africa" award by the Council on Tall Buildings and Urban Habitat.[158] On 28 September 2010 Burj Khalifa won the award for the best project of the year at the Middle East Architect Awards 2010.[159] Awards Chair Gordon Gill, of Adrian Smith + Gordon Gill Architecture, said:

> We are talking about a building here that has changed the landscape of what is possible in architecture – a building that became internationally recognized as an icon long before it was even completed. 'Building of the Century' was thought a more apt title for it.[160]

Burj Khalifa was also the recipient of the following awards.[161][162]

Year	Award
2012	Award of Merit for World Voices Sculpture, Burj Khalifa Lobby from Structural Engineers Association of Illinois (SEAOI), Chicago.
2011	Interior Architecture Award, Certificate of Merit from AIA – Chicago Chapter.
	Distinguished Building Award, Citation of Merit from AIA – Chicago Chapter.
	Interior Architecture Award: Special Recognition from AIA – Chicago Chapter.
	Design Excellence Award: Special Function Room.
	Excellence in Engineering from ASHRAE (American Society of Heating, Refrigerating, and Air-Conditioning Engineers) – Illinois Chapter.
	Outstanding Structure Award from International Association for Bridge and Structural Engineering.
	Decade of Design, Presidential Commendation in Corporate Space Small from International Interior Design Association (IIDA).
	Decade of Design • Best of Category/Mixed Use Buildings from International Interior Design Association (IIDA).
	GCC Technical Building Project of the Year from MEED (formerly *Middle East Economic Digest*).
	Project of the Year from MEED.
2010	International Architecture Award.
	Arab Achievement Award 2010: Best Architecture Project from Arab Investment Summit.
	Architecture Award (Mixed Use) Dubai from Arabian Property Awards.
	Architecture Award (Mixed Use) Arabian Region from Arabian Property Awards.
	International Architecture Award from Chicago Athenaeum.
	American Architecture Award from Chicago Athenaeum.
	Commercial / Mixed Use Built from Cityscape.
	Best Mixed Use Built Development in Cityscape Abu Dhabi.
	Skyscraper Award: Silver Medal from Emporis.
	Award for Commercial or Retail Structure from Institution of Structural Engineers.
	International Architecture Award (Mixed Use) from International Commercial Property Awards.
	Special Recognition for Technological Advancement from International Highrise Awards.
	Best Structural Design of the Year from LEAF Award.
	International Projects Category: Outstanding Project from National Council of Structural Engineers Associations.
	Best of What's New from Popular Science Magazine.
	Spark Awards, Silver Award.
	Excellence in Structural Engineering: Most Innovative Structure from SEAOI.

See also

- List of buildings in Dubai
- List of buildings with 100 floors or more
- List of development projects in Dubai
- List of tallest buildings and structures
- List of tallest freestanding structures
- List of tallest buildings in Dubai
- List of tallest buildings in the United Arab Emirates
- List of tallest buildings
- List of tallest structures
- Jeddah Tower

Notes

a. Arabic: بُرْج خَلِيفَة, *Burj Khalīfah*, pronounced [bʊrd͡ʒ xaˈliːfa], lit. 'Khalifa tower'

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